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Blending Gradient Boosting Trees on Alternative Targets for Fuel Consumption Estimation

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Abstract

This study addresses the problem of aircraft fuel consumption estimation using sparse and heterogeneous trajectory data derived from ADS-B and ACARS recordings. Accurate fuel prediction is essential to improve operational efficiency, reduce emissions, and support cost-effective flight planning; yet it remains a challenge due to missing data, irregular sampling, and high variability in fuel measurements. To address these challenges, we develop a machine learning framework based on Gradient Boosting Decision Trees, combining XGBoost and CatBoost models trained on alternative target representations. We investigate transformations of the target variable from raw fuel consumption to fuel flow and its logarithmic form, demonstrating that these representations significantly improve numerical stability and model performance by reducing variance and mitigating gradient imbalance. The proposed method achieves a Root Mean Square Error of 220.67 kg, securing fourth place in the PRC Data Challenge 2025. While the results demonstrate the effectiveness of target transformation and ensemble learning in handling noisy real-world flight data, they also reveal a performance ceiling driven by the absence of key physical variables such as take-off weight and environmental conditions, highlighting the importance of integrating data-driven methods with physically grounded features to further improve fuel consumption prediction accuracy.

Keywords: Fuel consumption estimation; ADS-B; ACARS; GBDT; Ensemble Learning

Abbreviations: GBDT: Gradient Boosting Decision Tree(s), ADS-B: Automatic Dependent Surveillance–Broadcast, ACARS: Aircraft Communications Addressing and Reporting System, RMSE: Root Mean Square Error

1. Introduction

The aviation industry faces increasing pressure to minimize its environmental footprint. Central to this challenge is the accurate estimation of aircraft fuel burn, which is essential for precise flight planning, emissions monitoring, and cost management. While traditional physical models offer a baseline, the growing availability of high-fidelity ADS-B and ACARS data provides an opportunity to leverage machine learning for more accurate, data-driven predictions.

In this paper, we present the methodology and results of our submission to the PRC Data Challenge 2025, where our model achieved fourth place. Our approach focuses on a coordinated GBDT framework using XGBoost and CatBoost. The key element lies in the exploration of alternative target representations. We demonstrate that the inherent biases of different target spaces can be effectively counterbalanced through a weighted ensemble. By blending CatBoost and XGBoost on different targets, we achieve a consensus that outperforms single-target architectures.

This paper is organized as follows. Section 2 introduces the competition and external datasets. Section 3 describes the data processing and feature engineering strategy. Section 4 outlines the model selection, hyperparameter tuning via Optuna, and ensemble architecture. Section 5 discusses the results and the identified performance bottlenecks. Section 6 concludes the study.

2. Dataset

2.1 Competition Data

The competition data provided by Sun et al. [1] consists of ADS-B and ACARS recordings of 15761 flights, spanning 27 different aircraft types of 3 Wake Turbulence Categories (WTC) in total, as shown in Table 1. Among these flights, 11037 are for training, 1888 are for preliminary ranking, and 2836 are reserved for final submission.

Table 1. Aircraft Distribution by ICAO WTC Code

WTC	Typecode	Flights	WTC	Typecode	Flights
H	A359	2196	M	A20N	5221
	A332	714		A320	4074
	B788	655		B738	753
	B789	320		A21N	507
	B77W	295		A321	325
	A333	212		A319	45
	B744	175		B737	36
	B772	133		B739	34
	A306	12		B38M	30
	B752	6		A318	1
	B763	4		B39M	1
	B748	3		B734	1
	B77L	3	J	A388	4
	MD11	1			

The raw trajectory data as is a fusion of ADS-B and ACARS data streams. Due to the nature of real-world data, both exhibit a degree of sparsity. Our approach addresses this by first attempting to cross-feed both data streams, using one source to cover the gaps present in the other wherever overlaps permit. In scenarios where both sources are simultaneously absent and a gap remains, we resort to the approach detailed in the feature engineering process, as further explained in section 3.

2.2 External Open Source Data

To enhance the predictive capability, external datasets are incorporated to supplement the primary trajectory information. A core component is the FAA database, which provides critical aircraft specifications and operational characteristics such as Maximum Takeoff Weight (MTOW), Maximum Landing Weight (MALW), and structural dimensions [2]. The variables considered from the FAA Database are:

- MTOW
- MALW
- Wingspan
- Length
- Tail height
- Wheelbase

- Approach Speed
- ICAO WTC
- Gear Configuration
- Engine Class

In addition to the FAA database, the engine parameters database present in OpenAP is considered to provide a more accurate estimate of fuel burn [3]. However, preliminary testing revealed that the inclusion of engine characteristics led to a degradation in overall model performance. This decline is likely attributable to a combination of feature redundancy. By introducing highly correlated variables already captured by the ICAO identifier and trajectory features, the model complexity increased without a proportional gain in information, leading to overfitting and reduced generalization.

3. Data Processing and Feature Engineering

3.1 Feature Generation and Proxies

The feature engineering strategy focuses on transforming trajectory data and static aircraft specifications into a feature matrix. Following the exclusion of engine parameters, the feature set focuses on kinematic summaries, aircraft geometry, and derived proxies to capture the underlying physics of fuel consumption.

The model aggregates trajectory data into summary statistics for each segment to capture the kinematic state. These features include mean altitude, Mach number, the mean vertical rate and the maximum absolute vertical rate. To account for environmental context, the process applies cyclic transformations to the takeoff hour, and incorporates the Great Circle distance between the origin and destination.

3.2 Synthetic Mass

Because instantaneous aircraft weight is not directly observable in the dataset, the feature engineering strategy relies on an approximate synthetic mass. Recognizing that fuel flow is physically dependent on weight, a linear time-dependent decay is implemented as shown in Equation 1. Note that this approximation uses MTOW as the initial weight, which is not necessarily the case for most flights in the dataset.

$$\hat{m} \approx \text{MTOW} \cdot \left(1.0 - \frac{t - t_{takeoff}}{t_{total}}\right) \tag{1}$$

This mass proxy is then interacted with kinematic variables to represent power demand and momentum. For instance, the Energy Rate Index serves as a proxy for the thrust required to change potential energy. Additionally, geometric features such as the Tail Length Ratio are included to allow the model to differentiate between the drag profiles of various airframes. The full list of all features is available on the GitHub repository [4].

3.3 Model Integration

The pipeline addresses intervals of missing trajectory data through an explicit binary flag named `is_missing_data`. In these instances, the system sets all corresponding kinematic values to 0.0, which allows the models to treat the absence of data as a distinct mode.

Categorical variables including aircraft type and wake turbulence category are processed according to the specific requirements of the model ensemble. For CatBoost, these variables remain in string

format to utilize native categorical split logic. For XGBoost, the features undergo one-hot encoding to ensure compatibility and maintain a uniform input dimensionality across the final ensemble.

4. Model Selection and Training

4.1 Choice of GBDT Models: XGBoost and CatBoost

The choice of machine learning framework for this task is primarily driven by the sparsity and noise present in ADS-B and ACARS data streams. Categorical features, such as aircraft type, also play an important role in prediction, making Gradient Boosting Decision Trees (GBDTs) particularly well-suited for this application. GBDTs iteratively build ensembles of decision trees, where each tree corrects the errors of its predecessors, enabling the model to capture complex nonlinear relationships and interactions in the data.

The suitability of tree-based models for fuel burn prediction is also supported by prior work. Horiguchi et al. [5] demonstrated that XGBoost significantly outperforms deep neural networks in operational settings, reducing prediction errors by nearly 40%. Furthermore, Baumann and Klingauf [6] highlighted that decision trees outperform neural networks in specific flight phases, and can be trained more efficiently.

Among GBDT implementations, we focus on XGBoost [7] and CatBoost [8] due to their complementary properties. XGBoost constructs asymmetric trees, allowing splits to occur independently across branches. This flexibility enables it to capture complex patterns and adapt to highly non-linear feature relationships. However, the same flexibility can make it more sensitive to noise and potentially prone to overfitting.

CatBoost on the other hand constructs symmetric (oblivious) trees, where the same split is applied across all branches at a given depth. This structure provides better regularization and results in more conservative predictions. As noted by Hancock and Khoshgoftaar [9], CatBoost is specifically designed for the heterogeneous and high-cardinality categorical data prevalent in big data tasks. CatBoost natively handles categorical features, which reduces the need for preprocessing and improves robustness.

These architectural differences lead to complementary prediction biases, and understanding them motivates the later exploration of model blending to improve overall predictive performance.

4.2 Evaluation Metric

The Root Mean Square Error (RMSE) serves as the primary optimization objective and evaluation metric, as dictated by the PRC Data Challenge requirements:

$$\text{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2} \tag{2}$$

where y_i represents the true fuel consumption and $\hat{y}_i$ is the model prediction for a given segment.

4.3 Alternative Target Transformations

The distribution of the target variable significantly impacts the training stability and generalization for GBDTs. As demonstrated by Zhang et al. [10], skewed or high-variance distributions in regression tasks often induce a "gradient imbalance," where a minority of high-magnitude samples disproportionately dictate the boosting process.

The empirical distribution of the data summarized in Table 2 clearly highlights this challenge: the Raw Fuel target exhibits a massive variance of 780675, spanning several orders of magnitude. This

motivates the exploration of alternative target transformations to improve the numerical stability of the boosting process, and model accuracy.

Table 2. Mean and variance of alternative fuel target transformations

Transformation	Mean	Variance
Raw Fuel	557.71	780675
Log Raw Fuel	5.5404	1.5734
Fuel Flow	0.9319	0.5791
Log Fuel Flow	0.5976	0.1129

First, we evaluate a physical normalization by deriving Fuel Flow (kg/s) using each segment duration. This transformation is motivated by the fact that fuel flow is a characteristic, bounded variable dictated by engine type and thrust setting. Moreover, predicting a rate rather than an absolute quantity decouples the target from the variable segment duration which contributes to reducing variance dramatically. This approach also aligns with OpenAP and Acropole implementations.

Second, we apply a logarithmic transformation to further reduce the variance of the target:

$$y = \ln(1 + x) \tag{3}$$

Where x is the fuel flow and y represents the transformed target. This compresses the range of values so that residuals from different flight regimes contribute more evenly to the objective function. As shown in Table 2, Log Raw Fuel reduces variance to 1.5734, while Log Fuel Flow achieves the minimum observed variance of 0.1129. By combining these two pathways, the Log Fuel Flow transformation prevents the GBDT from being dominated by peak consumption events, leading to better performance overall.

4.4 Hyperparameter Tuning

To maximize the performance of the gradient boosting estimators, we employ the Optuna framework for automated hyperparameter optimization. We use the Tree-structured Parzen Estimator (TPE) sampler to navigate the search spaces for both CatBoost and XGBoost over 30 trials each. This approach enables more efficient exploration than standard grid search by modeling the probability of parameter performance based on previous trial history, focusing the search on regions of the parameter space that yield the lowest objective values.

4.4.1 Cross-Validation and Leakage

We implement a 3-fold `GroupKFold` cross-validation, using the `flight_id` as the grouping constraint. This ensures that observations from the same flight are never split between the training and validation sets, preventing data leakage and providing a realistic estimate of performance on future flights. For computational efficiency during the optimization phase, the objective function evaluates the performance on the primary validation fold before proceeding to the next trial.

4.4.2 Model Configurations and Search Spaces

The search focuses on parameters governing tree depth, learning rates, and regularization to mitigate overfitting. While both models share similar boosting principles, their specific internal parameter names differ according to library-specific implementations. The search spaces and settings for both models are summarized in Table 3.

The tuning settings are tailored to each model, particularly regarding the iteration and estimator counts. For CatBoost, the `iterations` count is fixed at 2000, leveraging the model's internal overfit

detection to reduce computation time. This allows the optimization effort to concentrate on depth and regularization. In contrast, the XGBoost search space includes `n_estimators` as a range. This enables the TPE sampler to manage model complexity by navigating between the number of boosting rounds and the learning rate as a measure to prevent overfitting.

Table 3. Hyperparameter search spaces and internal settings used with Optuna

Parameter	Value / Range	Description
CatBoost		
`depth`	[6, 10]	Maximum tree depth
`learning_rate`	[0.02, 0.2]	Step size shrinkage (log-scale)
`l2_leaf_reg`	[1.0, 15.0]	L_2 regularization coefficient
`random_strength`	$[10^{-9}, 10.0]$	Log-scale noise for split scoring
`iterations`	2000	Number of boosting rounds
`border_count`	128	Numerical feature quantization
`subsample`	0.8	Row sampling fraction
XGBoost		
`max_depth`	[6, 12]	Maximum tree depth
`learning_rate`	[0.01, 0.2]	Step size shrinkage (log-scale)
`reg_lambda`	[0.01, 10.0]	L_2 regularization coefficient (log-scale)
`reg_alpha`	[0.01, 10.0]	L_1 regularization coefficient (log-scale)
`n_estimators`	[500, 2500]	Number of boosting rounds
`min_child_weight`	[1, 10]	Minimum sum needed for leaf nodes
`gamma`	[0.1, 5.0]	Minimum loss reduction for splits
`subsample`	0.8	Row sampling fraction
`colsample_bytree`	0.8	Column sampling fraction

4.4.3 Final Model Training

Following the tuning phase, we select the best-performing parameters for a final training pass on the complete dataset. To ensure the models reach convergence with the larger volume of training data, the iteration counts used for tuning are increased. For this round we increase the iterations by 50% for CatBoost and 20% for XGBoost. These increments are heuristically chosen, with the higher value in CatBoost reflecting the sensitivity of the log-transformed target, where small residuals represent larger absolute errors than in the linear model. While these values provide a functional baseline for the ensemble, they remain subject to further refinement. The configurations obtained after the re-training provide the final models used in the ensemble, shown in Table 4.

The final parameters reveal a stark difference in how each model optimized for its specific target. XGBoost converged on a high-depth, low-learning-rate configuration to likely manage the variance of the raw fuel mass target. Even with a search space up to 12, the model settled at a depth of 10. Combined with a high `gamma` (3.3478) and significant `reg_alpha` (0.9168), this indicates that strict pruning was required for stability against large residuals [7].

In contrast, CatBoost settled on a higher learning rate (0.0467). The compressed range of the log-transformed target reduced the risk of large gradient updates, allowing for a more aggressive learning approach. CatBoost was intentionally capped at 10 to account for the symmetric tree architecture, where complexity grows exponentially with depth, increasing the tendency to overfit [8]).

Table 4. Final model parameters for XGBoost and CatBoost

CatBoost		XGBoost	
Parameter	**Value**	**Parameter**	**Value**
`depth`	10	`max_depth`	10
`learning_rate`	0.0467	`learning_rate`	0.0101
`l2_leaf_reg`	8.5190	`reg_lambda`	0.0583
`random_strength`	1.58×10^{-9}	`reg_alpha`	0.9168
`iterations`	3000	`n_estimators`	2791
`border_count`	128	`min_child_weight`	10
`subsample`	0.8	`gamma`	3.3478

Excessive depth, especially in a compressed log-target space, risks the model isolating individual observations rather than capturing trends.

4.5 Ensemble Creation

The final ensemble consists of XGBoost predicting Raw Fuel and CatBoost predicting Log Fuel Flow in a weighted linear combination. This configuration is the result of initial tests on combinations of all four targets in Table 2, where added complexity did not yield noticeably better results than the individual models. Instead, we shift toward a targeted blend of two models that account for each other's shortcomings.

4.5.1 Ablation Study and Analysis

We conduct an ablation study to determine the optimal combination of two models under different target transformations. As shown in Table 5, switching to Fuel Flow reduces error compared to using raw mass. The Log Fuel Flow transformation performs best for both XGBoost (265.24 kg) and CatBoost (265.19 kg), confirming that the alternative targets are a successful modeling choice.

Table 5. Comparison of alternative targets and average residuals for XGBoost and CatBoost

Model / Target	RMSE (kg)	Avgerage Residual (kg)
XGBoost		
Raw Fuel	281.22	+3.11
Log Raw Fuel	279.89	-15.34
Fuel Flow	267.73	+3.69
Log Fuel Flow	265.24	-4.52
CatBoost		
Raw Fuel	272.79	+2.85
Log Raw Fuel	278.15	-18.21
Fuel Flow	265.20	+3.22
Log Fuel Flow	265.19	-4.88
Ensemble	**220.67**	**-2.48**

Looking at the residuals, Table 5 reveeals that models trained in linear space tend to overpredict, while those in log space tend to underpredict. Despite its higher individual RMSE, the XGBoost

Raw Fuel model remains in the ensemble. This choice is motivated by its superior performance in predicting extreme values observed during testing. Log-space models perform poorly at such values because the transformation compresses outliers too close to the average.

4.5.2 Bias Correction via Weighted Blending

The ensemble leverages these opposing error profiles to reach a more balanced prediction. While the CatBoost Log Fuel Flow model provides the most accurate baseline (residual: -4.88 kg), the XGBoost Raw Fuel model (residual: +3.11 kg) serves as a necessary counterbalance. Although the linear Fuel Flow model has a bias closer to zero (+3.69 kg), the Raw Fuel model is the preferred ensemble partner due to its specific sensitivity to absolute mass, which prevents the final output from underpredicting during high-consumption segments. We combine these using a weighted average:

$$\hat{y}_{final} = w_A \cdot (\exp(\hat{y}_{CB}) - 1) \cdot \Delta t + w_B \cdot \hat{y}_{XGB} \tag{4}$$

Through empirical tuning, the weights are set at $w_A = 0.70$ and $w_B = 0.30$. This combination centers the average residual at -2.48 kg and reduces the RMSE to **220.67 kg**.

5. Results and Discussion

5.1 Performance Ceiling

The ensemble achieves a final RMSE of 220.67 kg, securing 4th place in the competition. This result confirms that the coordinated GBDT framework effectively captures the relationships in flight data. However, the model hits a clear performance ceiling observed around 200 kg RMSE. A comparison with other approaches in this domain reveals that the current statistical approach lacks the structural depth required to reach higher accuracy.

5.2 Bias vs. RMSE

A central question in the ensemble design is whether a zero average residual serves as the optimal goal. Intuitively, pairing the two Fuel Flow models -which individually show lower RMSE- seems the most logical path for bias cancellation. However, empirical testing indicates that this combination, despite its theoretical balance, fails to reach the performance of the chosen configuration.

The selected blend instead pairs the Log Fuel Flow model with the Raw Fuel model because of observed error behavior at the distribution tails. While logarithmic transformations stabilize variance, they lead to significant underprediction in high-consumption segments. The Raw Fuel model, despite a higher individual RMSE and positive bias, maintains better performance in this region. This suggests that the ensemble's strength lies in the diversity of its error distribution where pairing complementary models reach a more robust consensus, rather than in zero bias.

5.3 Identified Bottlenecks

The inability to further reduce the RMSE is tied to specific methodological shortcomings. The most significant factor is the absence of a dedicated Takeoff Weight (TOW) estimation step. Because fuel consumption is fundamentally a function of aircraft mass, the current approach forces the model to infer mass from correlated secondary features or MTOW rather than treating it as a primary feature. Evidence from higher-performing architectures suggests that an explicit mass estimation step is a prerequisite for better accuracy.

Furthermore, the current feature set remains relatively lean. Interestingly, attempts to integrate engine constants from the OpenAP database do not yield the expected performance gains and, in

several iterations, significantly degrade the results. This suggests that static domain-specific features are insufficient on their own. The performance ceiling is likely dictated by the omission of dynamic environmental variables, such as localized wind data or air density. Without these physical realities, the model remains limited to purely statistical correlations. The architectural search is also restricted to XGBoost and CatBoost, leaving the potential benefits of LightGBM or more complex stacked generalizations unexplored. These factors combined create a floor that purely data-driven GBDT models cannot breach without a more nuanced integration of flight physics.

5.4 Limitations and Generalization Risks

The weighting and feature selection processes follow an exploratory path. While this approach is backed by the substantial dataset, it highlights a persistent gap between statistical correlation and physical causation. The failure of added engine parameters to improve the model suggest that the GBRT framework may have exhausted the utility of static aircraft constants within this feature space. It remains unclear whether the current performance ceiling results from a potential overfit to the training distribution, the absence of a dedicated TOW estimation step, or the omission of atmospheric context. Without these physical guardrails, the model's ability to generalize to edge cases or significantly different operational environments remains unverified. The results indicate that while a targeted ensemble of different transformations provides significant gains, reaching higher performance levels requires moving beyond static heuristics toward a more physically grounded representation of the flight environment.

6. Conclusion

In this paper, we demonstrate that target variable representation and ensemble blending have a significant impact on fuel consumption prediction quality in a real-world sparse data setting. While the fusion of ADS-B and ACARS data helped mitigate inherent sparsity, intervals of missing trajectory data required the model to treat the absence of information as a distinct operational mode via explicit binary flags. The addition of FAA aircraft database parameters generally improved prediction accuracy. In contrast, the further addition of engine parameters degraded performance.

The implementation of alternative target representations provided insight into the prediction process. The Log Fuel Flow transformation achieved the most stable baseline, reducing variance from over 780,000 to approximately 0.11. However, log-space models systematically underpredicted peak consumption. We attribute this to the fact that logarithmic transformations compress high-value outliers, causing residuals to contribute more evenly to the objective function and preventing the GBDT from capturing extreme consumption events.

The final ensemble architecture was obtained through an iterative process that combined physical reasoning with empirical exploration. By pairing a variance-stabilized CatBoost model (Log Fuel Flow) with a positively biased XGBoost model (Raw Fuel), we successfully counterbalanced opposing errors. Throughout the training process, we utilized Optuna for hyperparameter tuning, which revealed that the XGBoost component required strict pruning and high regularization to remain stable against the large residuals of the raw fuel target. The final weighted combination yielded an RMSE of 220.67 kg, securing fourth place in the competition.

Accurate mass estimation remains the primary hurdle for sub-200 RMSE values. Because the methodology relies on static statistical proxies such as estimating mass decay from MTOW, it lacks the precision of true weight observations. To breach this ceiling, future work requires more advanced, physically grounded feature engineering, specifically the integration of an explicit TOW estimation step and atmospheric variables. Furthermore, the exploration of alternative frameworks such as LightGBM remains essential for translating the alternative target method into practice.

Author contributions

- Emre Demirci: Conceptualization, Data Curation, Formal Analysis, Investigation, Methodology, Software, Validation, Visualization, Writing (Original Draft), Writing (Review and Editing)
- Egeberk Baruh Karabulut: Conceptualization, Data Curation, Investigation, Methodology, Visualization, Writing (Original Draft), Writing (Review and Editing)

Open data statement

Competition data was sourced from the PRC Data Challenge 2025 organizers, Sun et al. [1]. We have augmented this with the public FAA aircraft database at [2] .

Reproducibility statement

Code pertaining to the final competition model can be found at https://github.com/edmrc22/prc25mj. All code is built using Python 3.12 and contains no proprietary material. The ensemble uses the libraries `catboost` (1.3.1), `xgboost` (2.0.3), and `optuna` (v4.4) alongside the latest `scipy` release.

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